

SQL PRACTICE NOTES - PHYSICIAN & DEPARTMENT TABLES

2520030021

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STEP 1: CREATE PHYSICIAN TABLE

The screenshot shows the SQL Developer interface. On the left, the 'SCHEMAS' pane lists databases: bankdb, bookflow_db, bookflow_week2_db, college_yt, practice, sql_clg, and sys. The main editor window displays the following SQL script:

```
1 • CREATE DATABASE IF NOT EXISTS aggregate_db;
2 • USE aggregate_db;
3
4 • CREATE TABLE physician (
5     employeeid INT PRIMARY KEY,
6     name VARCHAR(100),
7     position VARCHAR(100),
8     ssn VARCHAR(20)
9 );
```

Below the editor, the 'Output' pane shows the 'Action Output' table:

#	Time	Action	Message
✓ 25	20:57:38	SELECT first_table.* FROM first_table LEF...	1 row(s) returned
✓ 26	20:58:26	SELECT * FROM class LEFT JOIN class_in...	5 row(s) returned
✗ 27	21:02:24	USE aggregate_db	Error Code: 1049.
✓ 28	21:02:27	CREATE DATABASE IF NOT EXISTS aggr...	1 row(s) affected
✓ 29	21:02:35	USE aggregate_db	0 row(s) affected
✓ 30	21:02:50	CREATE TABLE physician (employeeid ...	0 row(s) affected

STEP 2: INSERT DATA INTO PHYSICIAN

Navigator: WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7* x JOINS

Limit to 1000 rows

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_dlg
- sys

```

1 • CREATE DATABASE IF NOT EXISTS aggregate_db;
2 • USE aggregate_db;
3
4 • CREATE TABLE physician (
5     employeed INT PRIMARY KEY,
6     name VARCHAR(100),
7     position VARCHAR(100),
8     ssn VARCHAR(20)
9 );
10 • INSERT INTO physician VALUES
11 (1, 'Karuturi Sai Akhil', 'Staff Internist', '2520030021'),
12 (2, 'Elliot Reid', 'Attending Physician', '2222222222'),
13 (3, 'Christopher Turk', 'Surgical Attending Physician', '3333333333'),
14 (4, 'Percival Cox', 'Senior Attending Physician', '4444444444'),
15 (5, 'Bob Kelso', 'Head Chief of Medicine', '5555555555'),
16 (6, 'Todd Quinlan', 'Surgical Attending Physician', '6666666666'),
17 (7, 'John Wen', 'Surgical Attending Physician', '7777777777'),
18 (8, 'Keith Dudemeister', 'MD Resident', '8888888888'),
19 (9, 'Molly Clock', 'Attending Psychiatrist', '9999999999');

```

Administration Schemas

Information

No object selected

Output

Action Output

#	Time	Action	Message	Duration
26	20:58:26	SELECT * FROM class LEFT JOIN class_in...	5 row(s) returned	0.000 se
27	21:02:24	USE aggregate_db	Error Code: 1049. Unknown database 'aggr...	0.016 se
28	21:02:27	CREATE DATABASE IF NOT EXISTS aggr...	1 row(s) affected	0.016 se
29	21:02:35	USE aggregate_db	0 row(s) affected	0.000 se
30	21:02:50	CREATE TABLE physician (employeed ...	0 row(s) affected	0.094 se
31	21:04:44	INSERT INTO physician VALUES (1, 'Karut...	9 row(s) affected Records: 9 Duplicates: 0 ...	0.015 se

STEP 3: CREATE DEPARTMENT TABLE

Navigator: WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7* x JOINS

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

```

7      position VARCHAR(100),
8      ssn VARCHAR(20)
9  );
10 • INSERT INTO physician VALUES
11 (1, 'Karuturi Sai Akhil', 'Staff Internist', '2520030021'),
12 (2, 'Elliot Reid', 'Attending Physician', '2222222222'),
13 (3, 'Christopher Turk', 'Surgical Attending Physician', '3333333333'),
14 (4, 'Percival Cox', 'Senior Attending Physician', '4444444444'),
15 (5, 'Bob Kelso', 'Head Chief of Medicine', '5555555555'),
16 (6, 'Todd Quinlan', 'Surgical Attending Physician', '6666666666'),
17 (7, 'John Wen', 'Surgical Attending Physician', '7777777777'),
18 (8, 'Keith Dudemeister', 'MD Resident', '8888888888'),
19 (9, 'Molly Clock', 'Attending Psychiatrist', '9999999999');
20
21 • CREATE TABLE department (
22     departmentid INT PRIMARY KEY,
23     name VARCHAR(100),
24     head INT
25 );

```

Administration Schemas

Information

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration /
27	21:02:24	USE aggregate_db	Error Code: 1049. Unknown database 'aggr...	0.016 sec
28	21:02:27	CREATE DATABASE IF NOT EXISTS aggr...	1 row(s) affected	0.016 sec
29	21:02:35	USE aggregate_db	0 row(s) affected	0.000 sec
30	21:02:50	CREATE TABLE physician (employeeid ...	0 row(s) affected	0.094 sec
31	21:04:44	INSERT INTO physician VALUES (1, 'Karut...	9 row(s) affected Records: 9 Duplicates: 0 ...	0.015 sec
32	21:05:19	CREATE TABLE department (departme...	0 row(s) affected	0.047 sec

STEP 4: Insert Department Data

Navigator: Schemas

- Filter objects
- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_dlg
- sys

Administration Schemas

Information

No object selected

Object Info Session

WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7*

Limit to 1000 rows

```

12 (2, 'Elliot Reid', 'Attending Physician', '222222222'),
13 (3, 'Christopher Turk', 'Surgical Attending Physician', '333333333'),
14 (4, 'Percival Cox', 'Senior Attending Physician', '444444444'),
15 (5, 'Bob Kelso', 'Head Chief of Medicine', '555555555'),
16 (6, 'Todd Quinlan', 'Surgical Attending Physician', '666666666'),
17 (7, 'John Wen', 'Surgical Attending Physician', '777777777'),
18 (8, 'Keith Dudemeister', 'MD Resident', '888888888'),
19 (9, 'Molly Clock', 'Attending Psychiatrist', '999999999'),
20
21 CREATE TABLE department (
22     departmentid INT PRIMARY KEY,
23     name VARCHAR(100),
24     head INT
25 );
26
27 INSERT INTO department VALUES
28 (1, 'General Medicine', 4),
29 (2, 'Surgery', 7),
30 (3, 'Psychiatry', 9);

```

Output

Action Output

#	Time	Action	Message
28	21:02:27	CREATE DATABASE IF NOT EXISTS aggr...	1 row(s) affected
29	21:02:35	USE aggregate_db	0 row(s) affected
30	21:02:50	CREATE TABLE physician (employe...	0 row(s) affected
31	21:04:44	INSERT INTO physician VALUES (1, 'Karut...	9 row(s) affected Records: 9
32	21:05:19	CREATE TABLE department (departme...	0 row(s) affected
33	21:05:51	INSERT INTO department VALUES (1, 'Ge...	3 row(s) affected Records: 3

1. INNER JOIN

Navigator

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_dlg
- sys

Administration

Schemas

Information

No object selected

Object Info

Session

WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7* JOINS

Limit to 1000 rows

```

33 p.name AS Head_Physician
34 FROM department d
35 JOIN physician p
36 ON d.head = p.employeeid;
37 SELECT *
38 FROM physician
39 WHERE position = 'Surgical Attending Physician';

```

Result Grid

employeeid	name	position	ssn
3	Christopher Turk	Surgical Attending Physician	3333333333
6	Todd Quinlan	Surgical Attending Physician	6666666666
7	John Wen	Surgical Attending Physician	7777777777
NULL	NULL	NULL	NULL

physician 2 x

Output

Action Output

#	Time	Action	Message	Duration /
✓ 30	21:02:50	CREATE TABLE physician (employeeid ...	0 row(s) affected	0.094 sec
✓ 31	21:04:44	INSERT INTO physician VALUES (1, 'Karut...	9 row(s) affected Records: 9 Duplicates: 0 ...	0.015 sec
✓ 32	21:05:19	CREATE TABLE department (departme...	0 row(s) affected	0.047 sec
✓ 33	21:05:51	INSERT INTO department VALUES (1, 'Ge...	3 row(s) affected Records: 3 Duplicates: 0 ...	0.016 sec
✓ 34	21:06:22	SELECT d.name AS Department, p.na...	3 row(s) returned	0.000 sec
✓ 35	21:06:39	SELECT * FROM physician WHERE positio...	3 row(s) returned	0.000 sec

WHERE + LIKE

Navigator

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

Administration Schemas

Information

No object selected

Object Info Session

WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7* x JOI

37 • SELECT *

38 FROM physician

39 WHERE position = 'Surgical Attending Physician';

40

41 • SELECT *

42 FROM physician

43 WHERE name LIKE 'John%';

Result Grid

Filter Rows:

Edit: Export/Import:

	employeeid	name	position	ssn
▶	7	John Wen	Surgical Attending Physician	777777777
•	NULL	NULL	NULL	NULL

physician 3 x

Output

Action Output

#	Time	Action	Message
✓ 31	21:04:44	INSERT INTO physician VALUES (1, 'Karut...	9 row(s) affected Records: 9 Duplicat
✓ 32	21:05:19	CREATE TABLE department (departme...	0 row(s) affected
✓ 33	21:05:51	INSERT INTO department VALUES (1, 'Ge...	3 row(s) affected Records: 3 Duplicat
✓ 34	21:06:22	SELECT d.name AS Department, p.na...	3 row(s) returned
✓ 35	21:06:39	SELECT * FROM physician WHERE positio...	3 row(s) returned
✓ 36	21:07:14	SELECT * FROM physician WHERE name ...	1 row(s) returned

COUNT()

Navigator: WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7*

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

```

38 FROM physician
39 WHERE position = 'Surgical Attending Physician';
40
41 • SELECT *
42 FROM physician
43 • WHERE name LIKE 'John%'; SELECT COUNT(*) AS Total_Physician:
44 FROM physician;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Total_Physicians
9

Administration Schemas

Information

No object selected

Result 4 x

Output

Action Output

#	Time	Action	Message
32	21:05:19	CREATE TABLE department (departme...	0 row(s) affected
33	21:05:51	INSERT INTO department VALUES (1, 'Ge...	3 row(s) affected Records: 3
34	21:06:22	SELECT d.name AS Department, p.na...	3 row(s) returned
35	21:06:39	SELECT * FROM physician WHERE positio...	3 row(s) returned
36	21:07:14	SELECT * FROM physician WHERE name ...	1 row(s) returned
37	21:07:29	SELECT COUNT(*) AS Total_Physicians F...	1 row(s) returned

Object Info Session

COUNT DISTINCT

Navigator: WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7*

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

```

44 FROM physician;SELECT COUNT(DISTINCT position)
45 FROM physician;
46
47 SELECT position,
48     COUNT(*) AS Total_Employees
49 FROM physician
50 GROUP BY position;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Administration Schemas

Information

No object selected

Object Info Session

Result 6 x

Output

Action Output

#	Time	Action	Message
34	21:06:22	SELECT d.name AS Department, p.na...	3 row(s) returned
35	21:06:39	SELECT * FROM physician WHERE positio...	3 row(s) returned
36	21:07:14	SELECT * FROM physician WHERE name ...	1 row(s) returned
37	21:07:29	SELECT COUNT(*) AS Total_Physicians F...	1 row(s) returned
38	21:07:46	SELECT COUNT(DISTINCT position) FRO...	1 row(s) returned
39	21:08:38	SELECT position, COUNT(*) AS Total_...	7 row(s) returned

Group By + Having

Navigator: Schemas

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_cdg
- sys

WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7* x JC

```

50 GROUP BY position;
51
52 • SELECT position,
53     COUNT(*) AS Total_Employees
54 FROM physician
55 GROUP BY position
56 HAVING COUNT(*) > 1;

```

Result Grid

position	Total_Employees
Surgical Attending Physician	3

Administration Schemas

Information

No object selected

Object Info Session

Result 7 x

Output

Action Output

#	Time	Action	Message
✓ 35	21:06:39	SELECT * FROM physician WHERE positio...	3 row(s) returned
✓ 36	21:07:14	SELECT * FROM physician WHERE name ...	1 row(s) returned
✓ 37	21:07:29	SELECT COUNT(*) AS Total_Physicians F...	1 row(s) returned
✓ 38	21:07:46	SELECT COUNT(DISTINCT position) FRO...	1 row(s) returned
✓ 39	21:08:38	SELECT position, COUNT(*) AS Total_...	7 row(s) returned
✓ 40	21:13:14	SELECT position, COUNT(*) AS Total_...	1 row(s) returned

ORDER BY

SCHEMAS

Filter objects

bankdb

bookflow_db

bookflow_week2_db

college_yt

practice

sql_clg

sys

Administration

Schemas

Information

No object selected

Object Info

Session

Limit to 1000 rows

54 FROM physician

55 GROUP BY position

56 HAVING COUNT(*) > 1;

57 SELECT *

58 FROM physician

59 ORDER BY name ASC;

60

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Cont

	employeeid	name	position	ssn
▶	5	Bob Kelso	Head Chief of Medicine	555555555
	3	Christopher Turk	Surgical Attending Physician	333333333
	2	Elliot Reid	Attending Physician	222222222
	7	John Wen	Surgical Attending Physician	777777777
	1	Karuturi Sai Akhil	Staff Internist	2520030021
	8	Keith Dudemeister	MD Resident	888888888
	9	Molly Clock	Attending Psychiatrist	999999999
	4	Percival Cox	Senior Attending Physician	444444444
	6	Todd Quinlan	Surgical Attending Physician	666666666
*	NULL	NULL	NULL	NULL

physician 8 x

Output

Action Output

#	Time	Action	Message	Duration / 1
✓	36 21:07:14	SELECT * FROM physician WHERE name ...	1 row(s) returned	0.000 sec /
✓	37 21:07:29	SELECT COUNT(*) AS Total_Physicians F...	1 row(s) returned	0.015 sec /
✓	38 21:07:46	SELECT COUNT(DISTINCT position) FRO...	1 row(s) returned	0.000 sec /
✓	39 21:08:38	SELECT position, COUNT(*) AS Total_...	7 row(s) returned	0.000 sec /
✓	40 21:13:14	SELECT position, COUNT(*) AS Total_...	1 row(s) returned	0.000 sec /
✓	41 21:13:36	SELECT * FROM physician ORDER BY na...	9 row(s) returned	0.000 sec /

ORDER BY DESC

SCHEMAS

Filter objects

bankdb

bookflow_db

bookflow_week2_db

college_yt

practice

sql_clg

sys

Limit to 1000 rows

60

•

SELECT *

61

FROM physician

62

ORDER BY employeeid DESC;

63

64

•

SELECT *

65

FROM physician

66

ORDER BY employeeid DESC;

Result Grid

Filter Rows:

Edit:

Export/Import:

	employeeid	name	position	ssn
▶	9	Molly Clock	Attending Psychiatrist	9999999999
	8	Keith Dudemeister	MD Resident	8888888888
	7	John Wen	Surgical Attending Physician	7777777777
	6	Todd Quinlan	Surgical Attending Physician	6666666666
	5	Bob Kelso	Head Chief of Medicine	5555555555
	4	Percival Cox	Senior Attending Physician	4444444444
	3	Christopher Turk	Surgical Attending Physician	3333333333
	2	Elliot Reid	Attending Physician	2222222222
	1	Karuturi Sai Akhil	Staff Internist	2520030021
*	NULL	NULL	NULL	NULL

physician 10 ×

Output

Action Output

#	Time	Action	Message
✓ 38	21:07:46	SELECT COUNT(DISTINCT position) FRO...	1 row(s) returned
✓ 39	21:08:38	SELECT position, COUNT(*) AS Total_...	7 row(s) returned
✓ 40	21:13:14	SELECT position, COUNT(*) AS Total_...	1 row(s) returned
✓ 41	21:13:36	SELECT * FROM physician ORDER BY na...	9 row(s) returned
✓ 42	21:14:02	SELECT * FROM physician ORDER BY em...	9 row(s) returned
✓ 43	21:14:22	SELECT * FROM physician ORDER BY em...	9 row(s) returned

MAX()

Navigator

SCHEMAS

Filter objects

bankdb

bookflow_db

bookflow_week2_db

college_yt

practice

sql_clg

sys

Administration

Schemas

Information

No object selected

Object Info

Session

WEEK1*ddlWEEK2SQL File 5*SQL File 6*SQL File 7*JOINS

Limit to 1000 rows

Filter Rows:

Export:

Wrap Cell Content:

63

64 • SELECT *

65 FROM physician

66 ORDER BY employeeid DESC;

67

68 • SELECT MAX(employeeid)

69 FROM physician;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

MAX(employeeid)

9

Result 11 ×

Output

Action Output

#	Time	Action	Message	Dura
✓ 39	21:08:38	SELECT position, COUNT(*) AS Total_...	7 row(s) returned	0.000
✓ 40	21:13:14	SELECT position, COUNT(*) AS Total_...	1 row(s) returned	0.000
✓ 41	21:13:36	SELECT * FROM physician ORDER BY na...	9 row(s) returned	0.000
✓ 42	21:14:02	SELECT * FROM physician ORDER BY em...	9 row(s) returned	0.000
✓ 43	21:14:22	SELECT * FROM physician ORDER BY em...	9 row(s) returned	0.000
✓ 44	21:14:48	SELECT MAX(employeeid) FROM physicia...	1 row(s) returned	0.000